

generate a translucent effect on objects such as leaves. The newly introduced Viewport Manager allows you to use hardware shaders in DirectX mode, which will be helpful in creating content for games.

Animation

Animation has always been a tough job in 3ds max and animators invest huge amounts of their time in creating and rendering animations. Version 5 introduces some new and enhanced animation features. Lets begin with Bone Tools, which is a collection of time-saving tools for building and modifying skeletal structures (bones) at any stage. In previous versions, working with bones was a tedious and complicated process. These tools have been made more accessible by placing them under the Character menu.

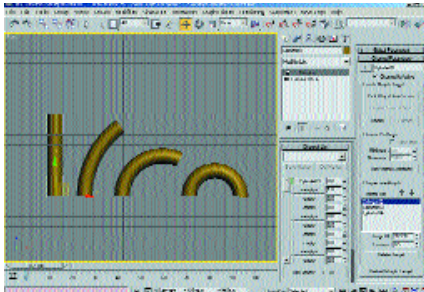
An all-new function named Skin Pose allows you to pose a particular object in a specific position and rotation (a transform position) and then store that posed state for later use as with the Skin modifier. Skin poses can be used for any object to store its current transforms for later use. For example, you can define a mesh as a character's skin pose and selectively return to that pose at any time during the animation process.

In the previous version, one had to use the Unhide option to detect Low Resolution Objects—a tedious process that can now be easily accomplished through the panel and you need not hide/unhide objects to do so.

Set Key Animation is a new animation mode, which supports pose-to-pose animation, especially for characters. Unlike in the previous version (4.26), artists can now set keys on only the necessary components of any character or animated object. Hence, animators can now use multiple animation methods

within 3ds max, allowing for a more interactive approach, without worrying about junk or unwanted key frames that generate unexpected results. There is also the Auto Key button, which can be used for generating keys automatically. It is equivalent to the Animate button in previous versions of 3ds max.

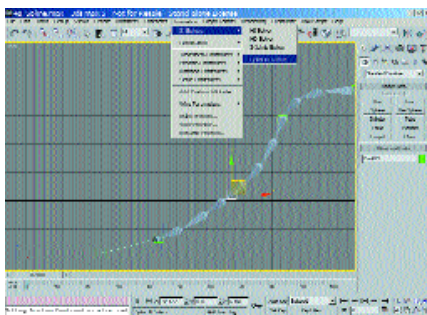
Morphing is commonly used for lip synchronisation and facial expressions on a 3D character, but can also be used to change the shape of any 3D model. The new Morpher Modifier supports Progressive Morphing, which lets you specify intermediate shapes, giving you more control over how the morph interpolation will take place.



Progressive morphing ensures smoother morphing of objects

Splines are object types such as lines, rectangles, circles, ellipses, arcs, stars, text and helixes. The new Spline IK Solver is an Inverse Kinematic (IK) solver that uses a spline as the basis for the shape of an IK chain—it provides a more flexible animation system than other IK solvers.

Spline IK is suitable for animating linked objects such as tails, snakes, tenta-

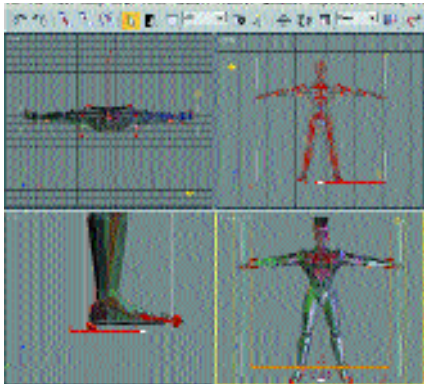


Spline IK Solver gives you greater control over animating bones

cles and ropes. This solver uses vertices of spline to animate, rather than individual bones. It uses a spline to determine the curvature of a series of bones or other linked objects. Since a spline is a curve, you can apply modifiers to it which allow you greater control over the bones as the vertices can be moved anywhere in 3D space and the linked structure can be shaped in any way.

Just like in the previous version, you can save and merge animation but while the older version only let you merge animations for one particular object, you can now merge an animation that is applied to one object with another. Hence, there is no need to repeat similar animation steps for another object. Moreover, the animation file that is being merged is still editable.

What also makes things easier for the animator is the Flex modifier. It can be used to simulate soft dynamics for an object: a ball falling from a height can thus show a characteristic wobble, making the animation more realistic.



The Skin Pose function makes character animation much simpler

Third-party Extensions to Boost your Productivity

Third-party extensions or plugins are a major source for adding extra modifications to the animation.

Face Station (www.eyematic.com) is a powerful application, which allows you to capture facial animation by capturing facial motion in real time in 3ds max itself. This program is powered by Eyematic's patented facial recognition technology, which quickly identifies facial features on a video stream and tracks them continuously. Thus the facial expressions and lip movements will be precisely reflected on your animated models.

Swift 3D max (www.swift3d.com) enables you to quickly render models and animations to a variety of vector file formats including Macromedia Flash (SWF), EPS, SVG and Adobe Illustrator. This is one of the most

versatile vector tools available for designers to create or repurpose 3D content for use with Macromedia's Flash. Thus, you can create amazingly small files while ensuring all the colours, lightning schemes and camera views are kept intact for animations.

If you want more audio tools inside 3ds max, you should consider using FlexSound (www.id8media.com). It allows you to add and layer unlimited number of WAV files to create entirely new composite sounds, which can consist of background noise, music, dialogue or even special effects within 3ds max itself. Once you have loaded the sound files, you can add and edit Flex Keys on the individual files from the Track View to compress or expand sections of the sound file for timing the existing animation.